

# Linear Algebra Foundations for Systems

Prepared by Staples Education

*Practice Set: Problems and Complete Solutions*

## Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb through linear combinations and bases, the matrix of a transformation, the determinant and its geometric meaning, the similarity transform, the characteristic equation and the trace/determinant shortcut, eigenvectors, the eigen-ansatz, and a capstone that solves a coupled system  $\mathbf{x}' = A\mathbf{x}$  end to end. Solutions follow in Part III.

1. **(Linear combination)** Compute the linear combination  $2(1, 0) + 3(0, 1)$ .
2. **(Basis test)** State the two conditions a set must meet to be a basis, then decide whether  $(1, 0)$  and  $(2, 0)$  form a basis of  $\mathbb{R}^2$ .
3. **(Matrix of a transformation)** For the matrix  $A$  with rows  $(1, 2)$  and  $(3, 4)$ , compute the image  $A\mathbf{x}$  of  $\mathbf{x} = (1, 1)$ .
4. **(Determinant, compute)** Find the determinant of the matrix with rows  $(3, 1)$  and  $(2, 4)$ , and state the factor by which it scales area.
5. **(Determinant, meaning)** Compute the determinant of the matrix with rows  $(1, 2)$  and  $(2, 4)$ . Is the matrix invertible? What does this say about its columns?
6. **(Change of basis)** Write the formula that expresses a matrix  $A$  in the basis given by a change-of-basis matrix  $P$ , and state what makes the result diagonal.
7. **(Characteristic equation)** Write the  $2 \times 2$  characteristic equation in terms of the trace and determinant, then find the eigenvalues of the matrix with rows  $(2, 1)$  and  $(1, 2)$ .
8. **(Eigenvectors)** For that same matrix (rows  $(2, 1)$ ,  $(1, 2)$ ), find an eigenvector for each eigenvalue.
9. **(Eigen-ansatz)** Show that substituting  $\mathbf{x} = e^{\lambda t}\mathbf{v}$  into  $\mathbf{x}' = A\mathbf{x}$  produces the eigenvalue equation  $A\mathbf{v} = \lambda\mathbf{v}$ .
10. **(Capstone — solve a system)** Using the eigenpairs from Problems 7–8, write the general solution of  $\mathbf{x}' = A\mathbf{x}$  for  $A$  with rows  $(2, 1)$  and  $(1, 2)$ , and classify the equilibrium at the origin.

## Part III — Complete Solutions

**Solution 1.** Scale each vector and add componentwise:

$$2(1, 0) + 3(0, 1) = (2, 0) + (0, 3) = (2, 3).$$

The scalars stay attached to their own directions; they do not merge into a single axis.

**Solution 2.** A basis must be *linearly independent* and must *span* the space. The vectors  $(1, 0)$  and  $(2, 0)$  are parallel, since  $(2, 0) = 2(1, 0)$ , so they are dependent and span only the  $x$ -axis:

$$\text{span}\{(1, 0), (2, 0)\} = \{(t, 0)\} \neq \mathbb{R}^2.$$

They are **not** a basis of  $\mathbb{R}^2$ .

**Solution 3.** Dot each row with the input  $\mathbf{x} = (1, 1)$ :

$$A\mathbf{x} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+2 \\ 3+4 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

**Solution 4.** Use  $\det = ad - bc$  with  $a = 3, b = 1, c = 2, d = 4$ :

$$\det A = (3)(4) - (1)(2) = 12 - 2 = 10.$$

The map scales every area by a factor of  $|\det A| = 10$ , and since  $\det A > 0$  it preserves orientation.

**Solution 5.** Compute the determinant:

$$\det \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = (1)(4) - (2)(2) = 4 - 4 = 0.$$

A zero determinant means the matrix is **not invertible**: it collapses the plane onto a line, so its columns are linearly dependent (indeed  $(2, 4) = 2(1, 2)$ ).

**Solution 6.** The similarity transform rewrites  $A$  in the new coordinates:

$$B = P^{-1}AP,$$

where the columns of  $P$  are the new basis vectors in the old coordinates. If those columns are eigenvectors of  $A$ , then  $B$  is **diagonal**, with the eigenvalues on the diagonal.

**Solution 7.** The characteristic equation is

$$\lambda^2 - (\text{tr } A)\lambda + \det A = 0.$$

For rows  $(2, 1), (1, 2)$ :  $\text{tr } A = 4$ ,  $\det A = 3$ , so

$$\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1) = 0 \Rightarrow \lambda = 3, 1.$$

**Solution 8.** Solve  $(A - \lambda I)\mathbf{v} = \mathbf{0}$  for each eigenvalue:

$$\lambda = 3: \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v}_1 = (1, 1);$$

$$\lambda = 1 : \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v}_2 = (1, -1).$$

**Solution 9.** Differentiate the trial  $\mathbf{x} = e^{\lambda t} \mathbf{v}$  (with  $\mathbf{v}$  constant) and substitute:

$$\mathbf{x}' = \lambda e^{\lambda t} \mathbf{v}, \quad A\mathbf{x} = e^{\lambda t} A\mathbf{v}.$$

Setting  $\mathbf{x}' = A\mathbf{x}$  and cancelling the nonzero scalar  $e^{\lambda t}$  leaves

$$\lambda \mathbf{v} = A\mathbf{v},$$

exactly the eigenvalue problem.

**Solution 10.** Each eigenpair contributes one exponential mode; superpose them:

$$\mathbf{x}(t) = c_1 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^t \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Both eigenvalues are real and positive, so every trajectory grows without bound: the origin is an **unstable node** (a source).

### Unit Key Takeaway

Every problem above is one geometric story: a matrix *moves* space. Its **columns** say where the basis lands, its **determinant** says how area is scaled and whether orientation flips, and its **eigenvectors** mark the directions it merely stretches. Feeding those eigenpairs into  $\mathbf{x} = e^{\lambda t} \mathbf{v}$  turns a coupled system into independent exponential modes — the exact machinery Unit 15 runs on.